

Replication Package

Breaking the Spiral of Silence

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1. Overview

This package reproduces every table and figure in the paper and its appendices. It contains data from surveys and experiments conducted by the authors with undergraduate students through the University of California, Berkeley's Experimental Social Science Laboratory (Xlab), two extension experiments, and analysis of the American National Election Studies (ANES) survey data.

All Stata analysis runs from one master file, `silence_master.do`. Figure A.1 is produced separately by a Jupyter notebook. Section 8 maps every exhibit to the code that produces it, the file it writes, and the file the manuscript reads.

2. Data availability

2.1 Summary

Dataset	Source	Provided here	Access
UC Berkeley Xlab experiments (baseline, midline, endline surveys; Zoom session rosters)	Collected by the authors via Qualtrics	Yes	—
Renaming committee extension experiment	Collected by the authors via Qualtrics	Yes	—
Polarization / extremity extension experiment	Collected by the authors via Qualtrics	Yes	—
American National Election Studies, 1974–2020	ANES	Extract only	Public

2.2 Data collected by the authors

All experimental data were collected by the authors using Qualtrics, with participants recruited from the UC Berkeley Xlab subject pool.

On “raw” versus “analysis” data. The raw data were collected via surveys on Qualtrics, so there is no methodological cleaning stage: the data as received from Qualtrics are already

one row per respondent, one column per item. The files in data/ are close to the Qualtrics exports, there were only two changes that were made between download and analysis of the Qualtrics data:

- **De-identification.** We removed identifiable information from every dataset that included them: ResponseId, IPAddress, LocationLatitude, LocationLongitude, email, first_name, last_name, login_id, twitter_handle, the Qualtrics panel fields RecipientEmail / RecipientFirstName / RecipientLastName / ExternalReference, and grantor. Every UC Berkeley Xlab (SONA) participant number appearing in the deposited data was mapped through a single global crosswalk to a new ID.
- **Renaming the variables.** See the codebook for detailed variable names and labels.

2.3 Third-party data

American National Election Studies. data/anes_full.csv and anes_full_inperson.csv are extracts aggregated to the year × topic level (448 rows each) from the ANES Time Series Cumulative Data File, covering 1974–2020.

3. Rights and license

- The authors certify that they have legitimate access to, and permission to use, all data employed in this manuscript.
- The authors certify that they have permission to redistribute and publish the data included in this package, and that any restrictions are documented in LICENSE.txt.

4. Contents

—	README.pdf	this file
—	LICENSE.txt	
—	silence_master.do	master file – runs all Stata analysis
—	Codebook.pdf	
—	data/	analysis datasets, de-identified (§2)
—	└ csv/	plain-text copies
—	Code/	one script per exhibit (§8)
—	Experiment/	Experimental documentation and IRB (§9)
—	└ experimental_design_randomization_sampling.pdf	
—	└ survey_instructions.pdf	
—	└ IRB_approval (4 pdf)	
—	Output/	all tables (.tex) and figures (.png)
—	logs/	run logs, written by the master file
—	Tempfiles/	intermediate files written during the run

4.1 Datasets

File	N	Used by
recruit_baseline_roster_TC.dta	454	Figure 3, Figure B.2
recruitment_baseline.dta	95	Table B.1
recruitment_baseline_2nd.dta	353	Figure E.11
first_movers_analysis.dta	50	Table 1, Tables C.5, C.6
TC_cleaned.dta	333	Tables 2, 3; Figures 4, 5; Tables C.7–C.11, D.12–D.16; Figures D.3–D.7
TC_cleaned_2nd.dta	201	Figure E.11, Tables E.21, E.22
TC_first_movers_merged.dta	383	Table C.2
attrition.dta	454	Tables C.3, C.4
extremity_first_movers_cleaned.dta	35	Table E.17
extremity_TC_merged.dta	123	Tables E.18, E.19
Berkeley_renaming.xlsx	254	Table E.20
anes_full.csv, anes_full_inperson.csv	448 each	Table E.23

We also provide a .csv copy of each .dta.

5. Computational requirements

5.1 Stata

Run with Stata 18 on macOS Sequoia

The following user-written commands are required:

Command	Package	Install
esttab, eststo, estadd	estout	ssc install estout
distinct	distinct	ssc install distinct
cibar	cibar	ssc install cibar
splitvallabels	labutil2	ssc install labutil2
heatplot	heatplot	ssc install heatplot
— (dependencies of heatplot)	palettes, colrspace	ssc install palettes / ssc install colrspace

fmm (Table D.15) and suest (Table 2) are Stata commands; fmm requires Stata 15 or later.

5.2 Python (Figure A.1 only)

Code/Figure A1.ipynb requires numpy, numba and matplotlib

6. Instructions to replicators

1. Download the package and keep the folder structure intact.
2. Open `silence_master.do` and edit the single path line near the top:

```
global userpath "/path/to/this/package"
```

The other globals (`$data`, `$output`, `$tempfile`, `$code`, `$logs`) derive from it and should not need editing.

3. Install the Stata commands listed in §5.1.
4. Run `silence_master.do`. All tables and figures are written to `Output/`; the full run is logged to `logs/silence_master.log`.
5. For Figure A.1, install the Python dependencies `numpy`, `numba` and `matplotlib`, then open `Code/Figure A1.ipynb` from the `Code/` directory and run all cells. It writes `Output/figure_A1.png` plus the four individual panels.

6.1 Output displayed rather than saved

Every exhibit writes a file to `Output/` except **Table D.15**, which reports finite-mixture estimates read off the Stata output. The master file captures it in `logs/Table_D15.log`, and `Code/Table D15.do` documents which part of that output feeds each cell of the table.

The other thing that appears in `logs/silence_master.log` rather than in a file:

- The SUEST rows of Table 2 — see §8.1.

7. Runtime

- `silence_master.do`: approximately 1 minute
- `Code/Figure A1.ipynb`: approximately 30 seconds

8. List of exhibits

Every exhibit traces through two stages inside this package: a script in `Code/` produces a file in `Output/`; that file is copied into the manuscript source (usually renamed, and where the paper shows a two-panel table, two `Output/` files are merged into one). The manuscript source and the compiled paper are deposited separately, so the last column below names the file as it appears there rather than a file included here.

Figures 1, 2, E.8, E.9 and E.10 are images of the experimental materials shown to participants and are not produced by code.

Exhibit	Code (Code/)	Produced file (Output/)	File in manuscript source
Figure 1	— <i>(experimental material)</i>	—	Graphs/aa_C.png, Graphs/aa_T.png
Figure 2	— <i>(experimental material)</i>	—	Graphs/exp_design.png
Figure 3	Figure 3.do	private_perceived_a.png	Graphs/private_perceived_a.png
Table 1	Table 1.do	ols_first_movers_expression.tex (A); logit_first_movers_expression.tex (B)	Tables/reg_first_movers_expression.tex
Figure 4	Figure 4.do	inference_midline_2.png	Graphs/inference_midline_2.png
Table 2	Table 2.do	ols_inference_tc_revised.tex + SUEST rows from the log	Tables/reg_inference_tc.tex
Table 3	Table 3.do	ols_expression_minority_tc.tex (A); ols_expression_majority_tc.tex (B)	Tables/reg_expression_tc.tex
Figure 5	Figure 5.do	inference_rename_schools.png;inference_death_penalty.png;inference_affirmative_action.png;inference_immunizations.png	same four files in Graphs/
Figure A.1	Figure A1.ipynb	figure_A1.png (+ four panel files)	Graphs/figure_A1.png
Table B.1	Table B1.do	appropriateness_summary.tex	Tables/appropriateness_summary.tex
Figure B.2	Figure B2.do	private_perceived_TC.png	Graphs/private_perceived_TC.png
Table C.2	Table C2.do	summary_ftc.tex	Tables/summary_ftc.tex
Table C.3	Table C3.do	attrition.tex	Tables/attrition.tex
Table C.4	Table C4.do	attrition_v2.tex	Tables/attrition_v2.tex
Table C.5	Table C5.do	summary_first_movers.tex	Tables/summary_first_movers.tex
Table C.6	Table C6.do	summary_first_movers_r.tex	Tables/summary_first_movers_r.tex
Table C.7	Table C7.do	logit_expression_minority_tc_topic.tex (A);	Tables/reg_expressionssion_topic.tex

Exhibit	Code (Code/)	Produced file (Output/)	File in manuscript source
		logit_expression_majority_ tc_topic.tex (B)	
Table C.8	Table C8.do	first_reduced_minority_tc.t ex (A); first_reduced_majority_tc.t ex (B)	Tables/2sls_first_reduced.tex
Table C.9	Table C9.do	2sls_expression_minority_ tc.tex (A); 2sls_expression_majority_t c.tex (B)	Tables/2sls_expression_tc.tex
Table C.10	Table C10.do	ols_inference_tc_did.tex	Tables/reg_inference_tcd_did.tex
Table C.11	Table C11.do	dist_private_silent.tex	Tables/dist_private_silent.tex
Figure D.3	Figure 5.do	inference_daylight_saving. png	Graphs/inference_daylight_savin g.png
Figure D.4	Figure D4.do	bar_express_dst.png	Graphs/bar_express_dst.png
Table D.12	Table D12.do	ols_inference_did.tex	Tables/reg_inference_did.tex
Table D.13	Table D13.do	ols_expression_minority_d id.tex (A); ols_expression_majority_di d.tex (B)	Tables/reg_expression_did.tex
Table D.14	Table D14.do	het_midline.tex (A); het_endline.tex (B)	Tables/het_demographics_v2.tex
Figure D.5	Figure D5.do	kdensity_rename_schools. png;kdensity_death_penalt y.png;kdensity_affirmative _action.png;kdensity_imm unizations.png	same four files in Graphs/
Table D.15	Table D15.do	<i>(none — see Logs/Table_D15.Log)</i>	Tables/ffm_midline.tex
Table D.16	Table D16.do	ols_expression_high_v.tex	Tables/reg_expression_high_v.te x
Figure D.6	Figure D6.do	bar_express_trans_athletes .png	Graphs/bar_express_trans_athlet es.png
Figure D.7	Figure D7.do	corr_heatmap.png	Graphs/corr_heatmap.png
Figure E.8	—	—	Graphs/Polarization_C.png,

Exhibit	Code (Code/)	Produced file (Output/)	File in manuscript source
	<i>(experimental material)</i>		_T.png
Table E.17	Table E17.do	reg_first_movers_exp.tex	Tables/reg_first_movers_exp.tex
Table E.18	Table E18.do	reg_tc_guesses_extremity.tex	Tables/reg_tc_guesses_extremity.tex
Table E.19	Table E19.do	reg_tc_exp_extremity.tex	Tables/reg_tc_exp_extremity.tex
Figure E.9	— <i>(experimental material)</i>	—	Graphs/Berkeley_Renaming_C.png, _T.png, _P.png
Table E.20	Table E20.do	reg_renaming.tex	Tables/reg_renaming.tex
Figure E.10	— <i>(experimental material)</i>	—	Graphs/New_Control.png, Graphs/New_Treatment.png
Figure E.11	Figure E11.do	inference_rename_schools_2025.png; inference_death_penalty_2025.png; inference_affirmative_action_2025.png; inference_marijuana_2025.png	same four files in Graphs/
Table E.21	Table E21.do	ols_inference_tc_2025.tex	Tables/reg_inference_tc_2025.tex
Table E.22	Table E22.do	ols_expression_minority_tc_2025.tex (A); ols_expression_majority_tc_2025.tex (B)	Tables/reg_expression_tc_2025.tex
Table E.23	Table E23.do	reg_anes_all.tex (A); reg_anes_inperson.tex (B)	Tables/reg_anes.tex

8.1 Notes on the mapping

- The SUEST row in Table 2— $\chi^2(1)$ statistics 10.54, 10.55, 8.26 with p-values 0.0012, 0.0012, 0.0040 — is displayed in the Stata log only and transcribed by hand; Table 2.do runs the suest tests but does not write it to Output/. The three values have been verified against the data and are correct as computed.
- **Table D.15** is the only exhibit with no file in Output/; see §6.1.

- **Panel merges.** Tables 1, 3, C.7, C.8, C.9, D.13, D.14, E.22 and E.23 each combine two Output/ files into a single file in the manuscript.
- **Renaming.** Some draft filenames differ from the Output/ filenames they come from (ols_inference_tc_revised.tex → reg_inference_tc.tex).

9. Human subjects and experimental documentation

This project was approved by the Institutional Review Board at Harvard University (**IRB22-0757**) and at the University of California, Berkeley (**2022-11-15819** and **2023-08-16614**). The experiment was pre-registered in the AEA RCT Registry as **AEARCTR-0009505**.

We also include the following experimental documentations as separate PDFs:

- Experimental design document + Participant recruitment, selection and eligibility criteria
- Survey instructions
- IRB approval letters from both institutions

10. Data citations

American National Election Studies. 2022. *ANES Time Series Cumulative Data File (1948–2020)* [dataset and documentation]. September 16, 2022 version. www.electionstudies.org

Ho, Yuen, and Yihong Huang. 2026. "Data and Code for: Breaking the Spiral of Silence."